

TCL Commands : → (AUTO COMMIT)

- (i) Commit (save full)
 - (ii) Rollback (back)
 - (iii) Savepoint (save till here)
- To use them we must use START TRANSACTION;

These commands are mainly used to control transactions of the DML → Data Manipulation Commands.

Normalization : → It is like organising a messy room. Instead of throwing everything in one giant pile, we put things in the right drawers & shelves, so nothing is duplicated & everything is easy to find.

Problem it Solves : → Imagine we have an excel sheet (large) tracking orders.

Order ID	C-Name	C-Phone	Product	Category	Price
1	Akash	9876543210	iphone	Electronics	8000
2	Akash	9876543210	ipad	Electronics	5000
3	Punita	1234567890	iphone	Electronics	8000

Problems : → (i) Akash's phone number is repeated — twice — if he changes his no how many places do we need to update?

(ii) "Electronics" & "iphone" are repeated — wasteful & error-prone.

These problems are eliminated by using Normalization.

The NORMAL FORM RULES : → (NF)

→ [Levels of Tidiness]

(i) 1NF → 1st Normal Form :
No messy lists in a cell.
Each cell should have one single value not like [iphone, ipad]

(ii) 2NF → 2nd Normal Form
No Partial Dependencies
If Product Price depends only on the product, create separate table.

(iii) 3NF → 3rd Normal Form
No chain dependencies
If the category depends on the product, & not on the order, it should be inside products table.

The Final Fix Using Normalisation

Customers Table

Customer ID	Name	Phone
C1	Akash	9876543210
C2	Punita	1234567890

Products Table

Product ID	Name	Category	Price
P1	iphone	Electronics	8000
P2	ipad	Electronics	6000

Orders Table

Order ID	Customer ID	Product ID
1	C1	P1
2	C1	P2
3	C2	P1

Window Functions : → AVG of ALL students

GROUP BY ⇒ Group on certain condition
* { clubbing into one }

WINDOW FUNCTIONS ⇒ They group by alongside
{ clubbing + individual } Show individual data
both AVG of All + 1 individual

{ WINDOW FUNCTION + QUERY }
{ BCNF + MINI-PROJECT }